

VERDENT

YouTube API Services Compliance

Step-by-Step Demonstration

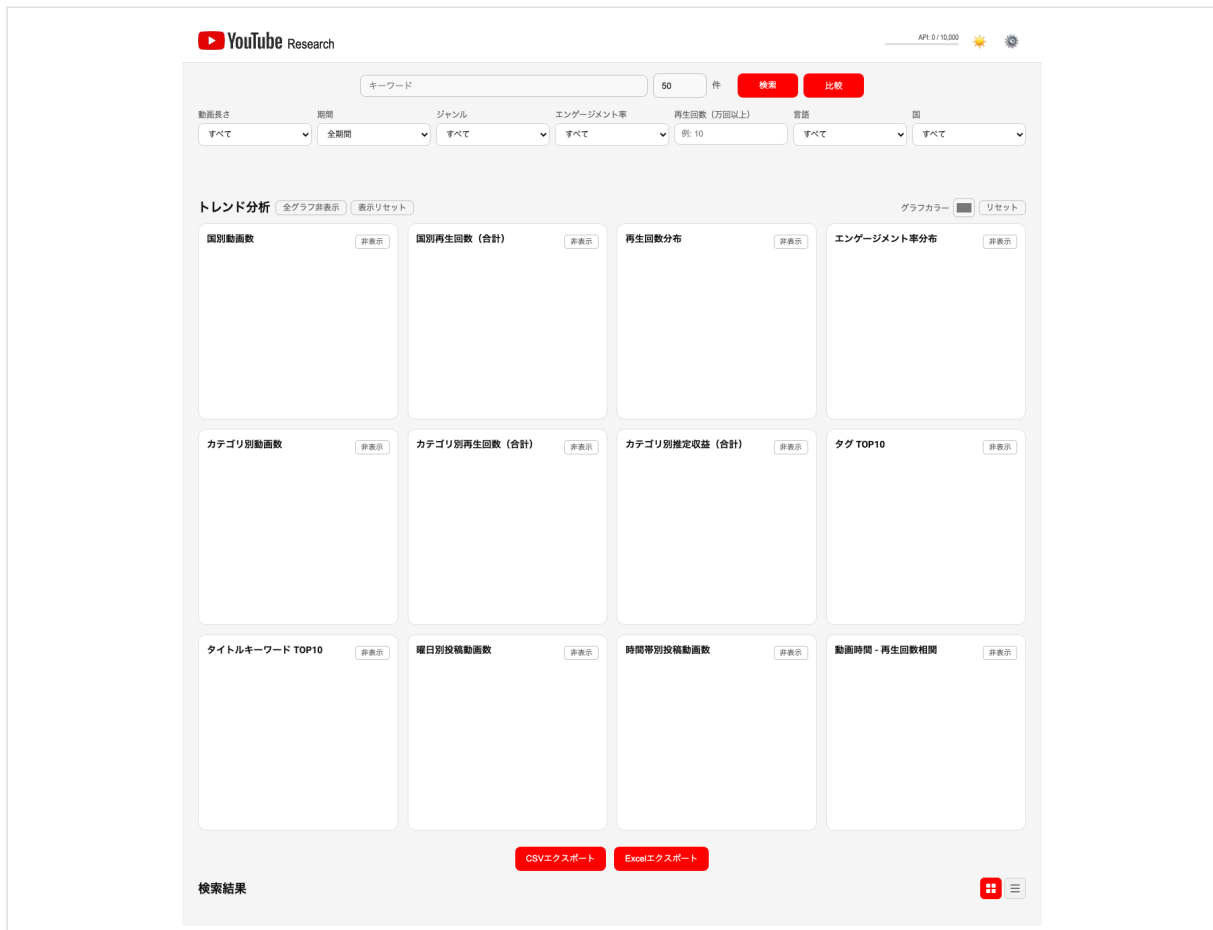
Application Name:	Verdent — YouTube Video Search & Analysis Tool
API Product:	YouTube Data API v3
Demonstration:	Live application screenshots with step-by-step annotations
Report Date:	June 2026
Contact:	developer@verdent.app

This report presents a live, step-by-step demonstration of the Verdent application, showing how it accesses and displays data from the YouTube Data API v3 in accordance with YouTube API Services Terms of Service and Developer Policies. Each page contains an actual screenshot from the running application, accompanied by an English description of the functionality demonstrated.

Step	Description	Screenshot File
1	Application Homepage — Search Interface	step1_homepage.png
2	Keyword Input — Entering 'gaming'	step2_search_input.png
3	TOP 10 Results — YouTube search.list Response	step3_top10_results.png
4	Trend Analysis — Statistical Graphs	step4_trend_analysis.png
5	Video Cards — Detailed Result Display	step5_card_results.png
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7	API Quota — Real-Time Usage Monitoring	step7_api_quota.png
—	Summary — API Endpoints & Compliance Notes	(text)

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Application Homepage — Search Interface

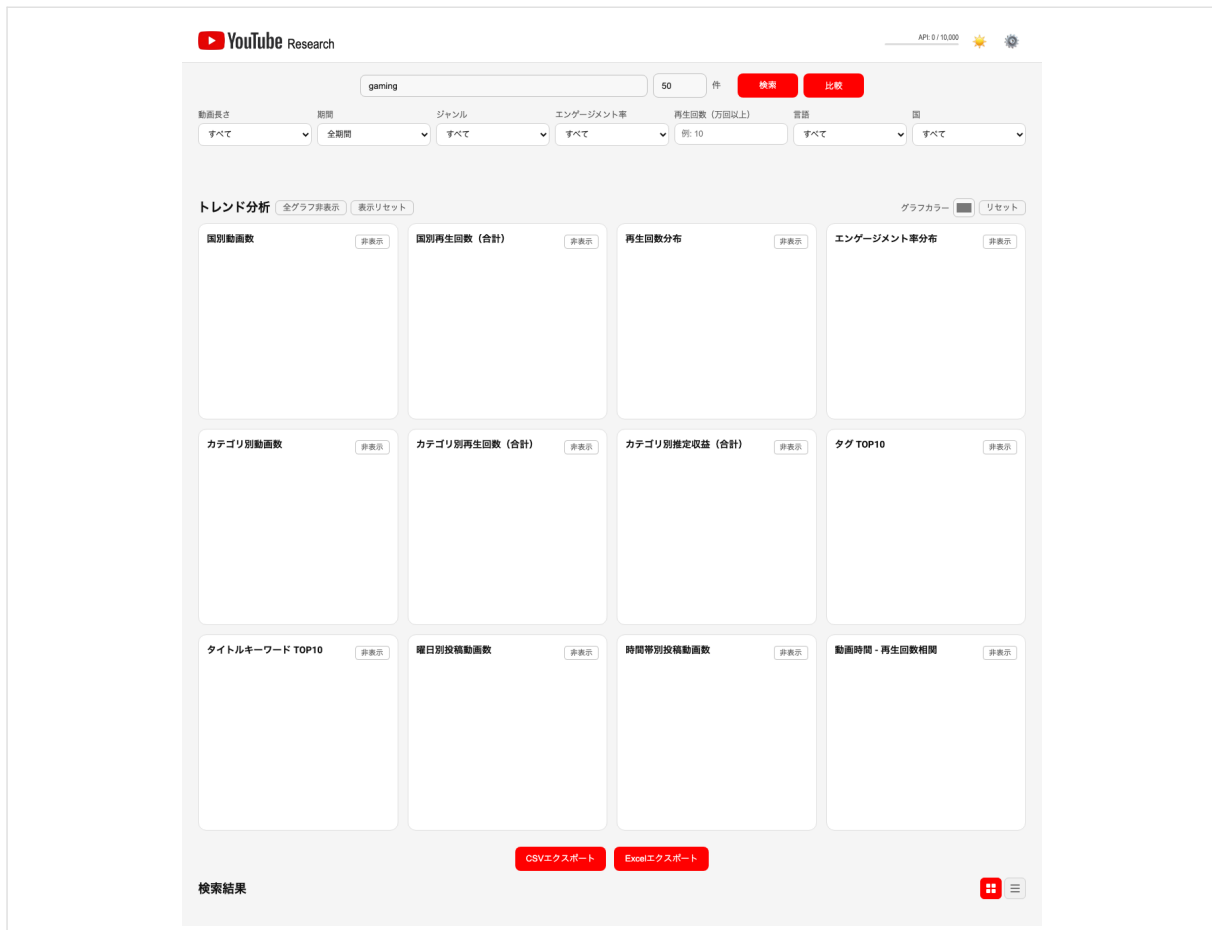


Screenshot: step1_homepage.png | Browser viewport: 1280 × 800 px | Captured with Playwright (headless Chromium)

The application loads at **http://localhost:8000/** and presents a clean search interface built with HTML, CSS, and vanilla JavaScript. The page includes a keyword input field, a result count selector, a search button, and a series of filter controls: video duration (all / short / normal), publication date range, content category, engagement rate, minimum view count, language, and region. A quota usage widget in the top-right corner displays real-time API quota consumption against the daily limit of 10,000 units. No YouTube API call is made until the user explicitly clicks the Search button, ensuring zero idle quota usage.

- Front-end: static HTML/CSS/JS served by FastAPI's StaticFiles middleware.
- Back-end: FastAPI (Python) running on uvicorn at port 8000.
- No user authentication is required; the app uses a server-side API key.
- All YouTube data is fetched server-side; the API key is never exposed to the browser.

2 Keyword Input — Entering 'gaming'

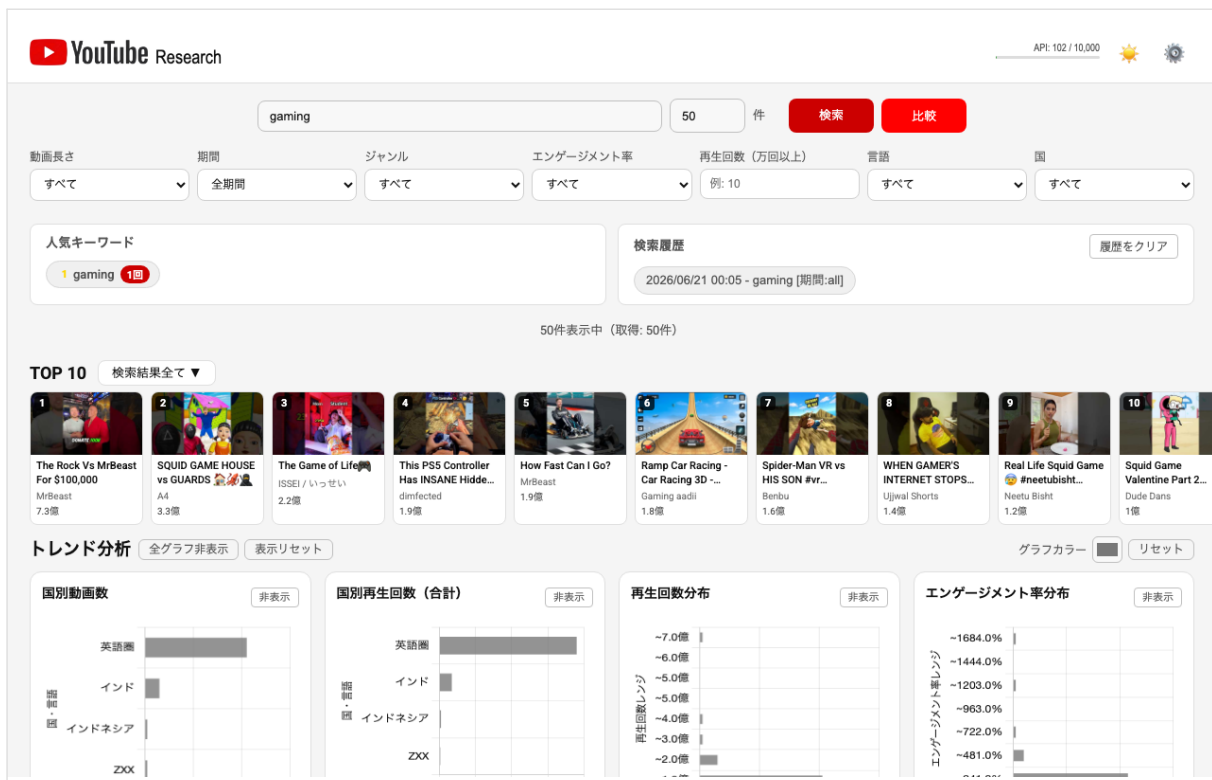


Screenshot: step2_search_input.png | Browser viewport: 1280 × 800 px | Captured with Playwright (headless Chromium)

The user types the keyword **'gaming'** into the search input field. At this stage, no API request has been sent yet — the application waits for the user to click the Search button. The search form allows users to specify a keyword, maximum number of results (1–200), and optional filters. A comparison mode (side-by-side keyword analysis) is also available via the 'Compare' button. Keyword search history is stored locally in the browser's `localStorage` only, and is never transmitted to the server or any third party.

- Keyword: 'gaming' (entered by the user).
- Optional filters: duration, date range, category, engagement, views, language, region.
- No API call is triggered until the Search button is clicked.
- Search history is stored only in browser `localStorage` — not on the server.

3 TOP 10 Results — YouTube search.list Response

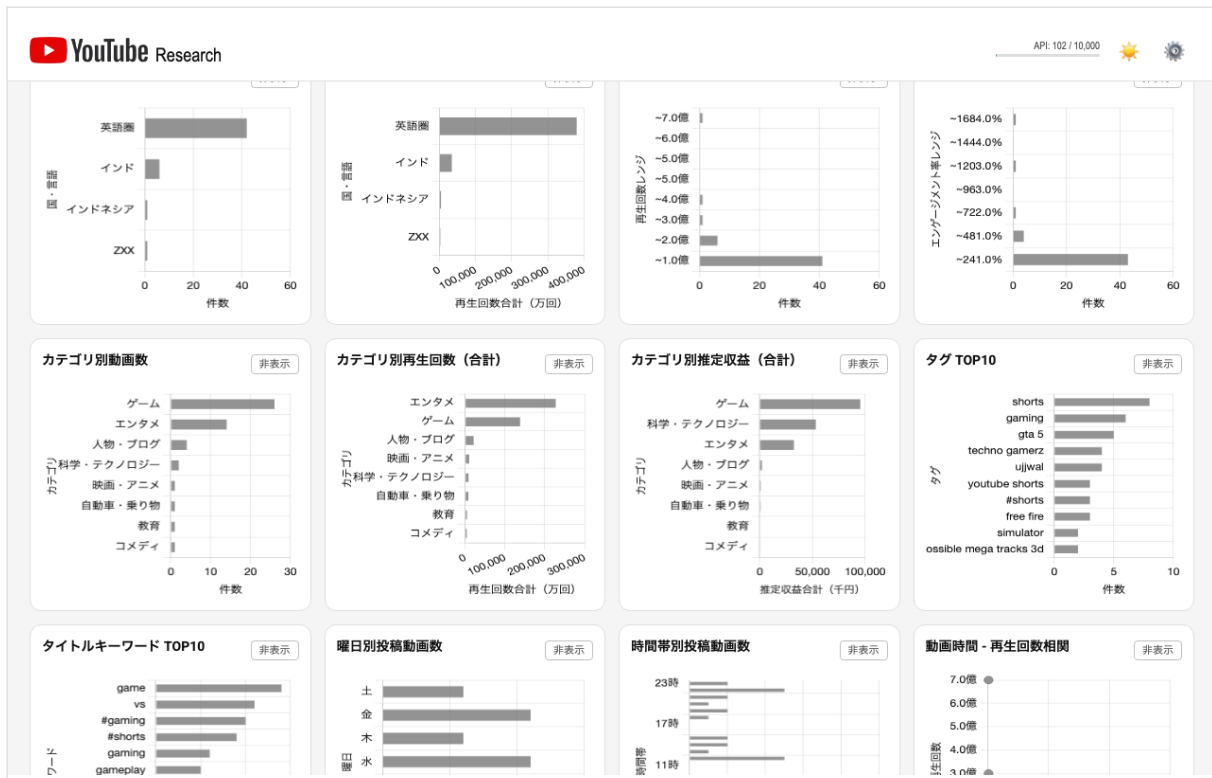


Screenshot: step3_top10_results.png | Browser viewport: 1280 × 800 px | Captured with Playwright (headless Chromium)

After the user clicks Search, the application calls the back-end **/api/search** endpoint. The back-end issues a **search.list** request to the YouTube Data API v3 (consuming 100 quota units) to retrieve video IDs and snippet metadata. It then batches the IDs and calls **videos.list** (1 quota unit per batch of up to 50) to fetch full statistics, and **channels.list** (1 unit per batch) for subscriber counts. The top 10 videos by view count are displayed in a prominently styled 'TOP 10' section. Each entry shows: video rank, thumbnail, title, channel name, subscriber count, view count, like count, duration, estimated revenue, and engagement rate.

- API call 1: **search.list** — **part=snippet, type=video, maxResults=50, order=viewCount** (100 quota units).
- API call 2: **videos.list** — **part=statistics, snippet, contentDetails** (1 unit per batch).
- API call 3: **channels.list** — **part=statistics, snippet** (1 unit per batch).
- Results are sorted by view count (descending) server-side.
- Results cached in-memory for 30 minutes (CACHE_TTL = 1800 s).

4 Trend Analysis — Statistical Graphs (Horizontal Bar Charts)



Screenshot: step4_trend_analysis.png | Browser viewport: 1280 × 800 px | Captured with Playwright (headless Chromium)

Below the TOP 10 section, Verdent renders a suite of statistical charts derived exclusively from the YouTube Data API response. All graphs use horizontal bar chart layout (Chart.js `indexAxis: 'y'`), making category labels easy to read. Charts displayed include: country-wise video count, country-wise total views, view count distribution, engagement rate distribution, category-wise video count, category-wise total views, category-wise estimated revenue, top-10 tags, top-10 title keywords, day-of-week posting distribution, hourly posting distribution, and video duration vs. view count correlation. All chart data is computed client-side from the API response — no additional API calls are made to render the analytics.

- All charts use Chart.js v4 rendered on elements.
- Data source: API fields returned by `search.list`, `videos.list`, `channels.list`.
- No additional API calls are triggered to render the trend analysis section.
- Individual charts can be shown/hidden; graph color is user-configurable.

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Video Cards — Detailed Result Display

YouTube Research API: 102 / 10,000

登録者数: 177,000 · エンゲージメント率: 312.7% ・ 長さ: 0分45秒 Top 3 Funniest Horror Game Screams 🤩 ... タグ: Top 3 Funniest Horror Game Screams (Bro got a...	登録者数: 3,970,000 · エンゲージメント率: 13.9% ・ 長さ: 0分29秒 タグなし	・ 長さ: 25分37秒 I Tried "Gaming" In EVERY Plane Seat... タグ: dantic, gaming, gaming setup, i tried, the, gamin...	登録者数: 45,700,000 · エンゲージメント率: 1.1% ・ 長さ: 0分53秒 Twitter: https://twitter.com/total_g... タグ: free fire, garena free fire, total gaming
Survive 99 Nights, Win \$99,000 MrBeast Gaming / 公開日: 2026/2/1 2:00:01 / ジャンル: ゲーム 再生回数: 4958.4万 · いいね: 592,368 · コメント: 26,266 推定収益: ¥991.7万 ※参考値 登録者数: 57,500,000 · エンゲージメント率: 0.9% ・ 長さ: 21分57秒 I still can't believe how close this was... タグなし	I Survived 100 Days In Realistic Minecraft MrBeast Gaming / 公開日: 2026/6/7 1:00:01 / ジャンル: ゲーム 再生回数: 4846.6万 · いいね: 842,089 · コメント: 24,186 推定収益: ¥969.3万 ※参考値 登録者数: 57,500,000 · エンゲージメント率: 0.8% ・ 長さ: 26分49秒 Me and the boys attempted to survive 100... タグなし	I BUILD MY NEW GAMING ROOM Techno / 公開日: 2023/7/16 3:27:15 / ジャンル: エンタメ 再生回数: 4820.6万 · いいね: 1,175,666 · コメント: 183,135 推定収益: ¥241万 ※参考値 登録者数: 5,420,000 · エンゲージメント率: 8.9% ・ 長さ: 21分18秒 I BUILD MY NEW GAMING ROOM #Gaming #Tec... タグ: my new gaming room, setup tour, gaming, hous...	Video Game glitches be like... pt. 5 #gaming #shorts NicsNite / 公開日: 2023/1/19 6:25:19 / ジャンル: ゲーム 再生回数: 4703.7万 · いいね: 2,427,328 · コメント: 20,573 推定収益: ¥477万 ※参考値 登録者数: 4,600,000 · エンゲージメント率: 10.2% ・ 長さ: 0分18秒 タグなし
naruto escape	naruto escape	naruto escape	1 MINUTE 10 YEAR

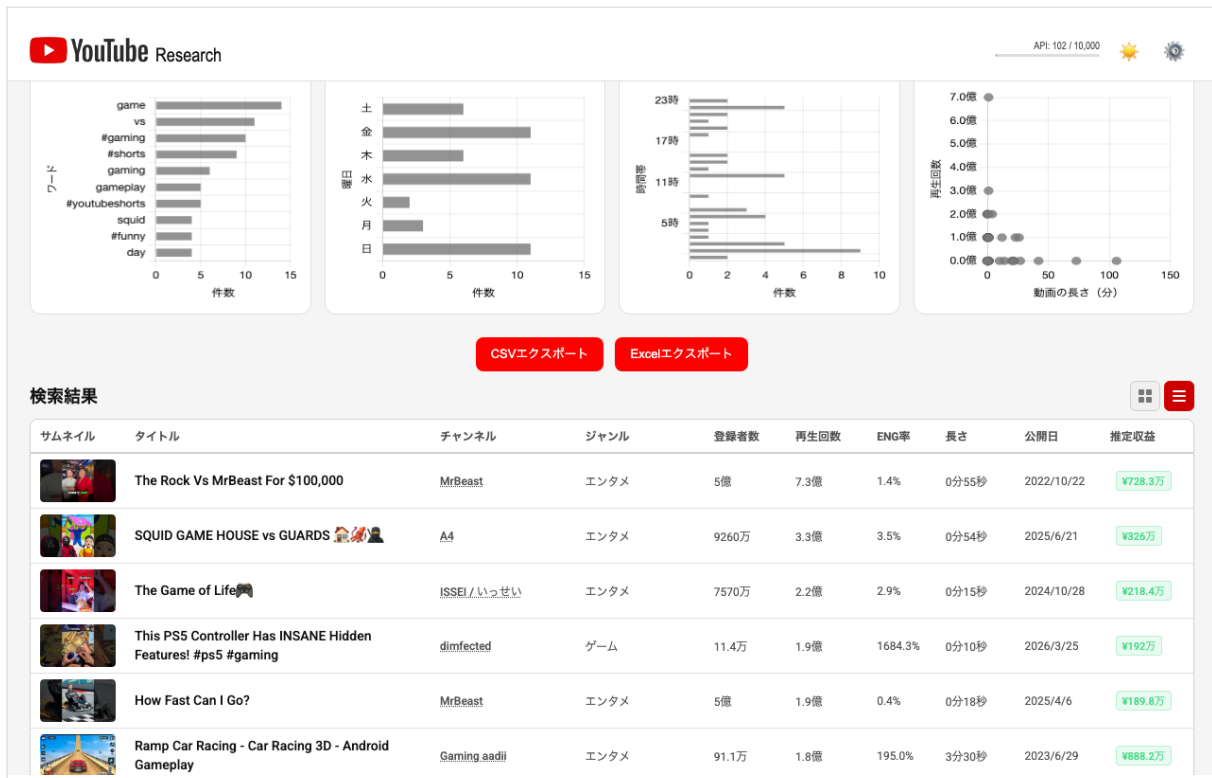
Screenshot: step5_card_results.png | Browser viewport: 1280 × 800 px | Captured with Playwright (headless Chromium)

The full search results list is displayed below the trend analysis section in a responsive card layout. Each video card shows: the video thumbnail (hotlinked from i.ytimg.com as provided by the API), the video title (clickable, opens a preview modal), the channel name (clickable, opens channel details modal), subscriber count, total view count, like count, video duration, publication date (displayed as relative age, e.g. '3 days ago'), content category, engagement rate (%), and estimated revenue (JPY). Each card also features a direct 'Open on YouTube' link and a Favourite button. Engagement Rate is calculated as $\text{likeCount} / \text{viewCount} \times 100$. Estimated Revenue uses a per-genre CPM (configurable by the user in Settings) multiplied by $\text{viewCount} / 1000$.

- Thumbnail: served from i.ytimg.com (YouTube CDN) — URL from snippet.thumbnails.
- Title / channel name: from snippet.title / snippet.channelTitle.
- View / like / comment counts: from statistics.*.
- Duration: parsed from contentDetails.duration (ISO 8601).
- Engagement Rate & Estimated Revenue: derived metrics, clearly labelled.
- Each card links back to <https://www.youtube.com/watch?v=>.

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List View — Tabular Data Comparison



Screenshot: step6_list_results.png | Browser viewport: 1280 × 800 px | Captured with Playwright (headless Chromium)

By clicking the list-view toggle button (the three-line icon next to the card-view icon), the user switches from the card layout to a compact tabular view. The table presents the same data fields as the card view — title, channel, subscribers, views, likes, duration, engagement rate, estimated revenue, and publication date — in a horizontally scrollable table. This view is particularly useful for quickly comparing multiple videos side-by-side without scrolling through individual cards. Switching view modes does not trigger any new API calls; the data is already in memory from the initial search.

- Toggle button: #view-list-btn (SVG icon, no text label).
- Table is horizontally scrollable on narrow viewports.
- No additional API calls are made when switching view modes.
- CSV and Excel export buttons are available above the table.

```
Pretty-print ☐  
{ "used": 102, "limit": 10000, "date": "2026-06-20" }
```

Screenshot: *step7_api_quota.png* | Browser viewport: 1280 × 800 px | Captured with Playwright (headless Chromium)

The **/api/quota** endpoint returns a JSON object showing the current API quota consumption tracked by the application. The response includes the number of quota units used so far in the current day, the daily quota limit (10,000 units by default), the number of remaining units, and a breakdown of calls made by endpoint type. This endpoint is also consumed by the quota widget in the UI header, which updates automatically after each search. If the remaining quota falls below 20% of the daily limit, the application displays a warning banner and rate-limits new requests. The quota counter is stored in-memory and resets at midnight UTC.

- Endpoint: GET /api/quota — returns JSON (no authentication required).
- Quota tracking: in-memory counter, incremented by `youtube_client.py` after each API call.
- Daily limit: 10,000 units (standard YouTube Data API v3 default quota).
- `search.list`: 100 units per call; `videos.list` / `channels.list`: 1 unit per batch.
- Counter resets at server restart (stateless, no persistent storage of quota data).

Summary — API Usage & Compliance

YouTube Data API v3 Endpoints Used

Endpoint	Parameters	Quota Cost	Purpose
search.list	part=snippet, type=video, q=<keyword>, maxResults=50, order=viewCount	100 units/call	Retrieve video IDs and snippet metadata for search results
videos.list	part=statistics,snippet,contentDetails, id=<comma-separated IDs>	1 unit/ID (Up to 50 IDs)	Retrieve view count, like count, comment count, and video details
channels.list	part=statistics,snippet, id=<comma-separated channel IDs>	1 unit/ID (Up to 50 IDs)	Retrieve subscriber count, channel thumbnail, and channel details

Caching Strategy (30-Minute In-Memory TTL)

Verdent implements a server-side in-memory cache keyed by a MD5 hash of the search parameters (keyword + max_results + duration_filter + published_after + category_id + language + region). Cache entries expire after **1,800 seconds (30 minutes)**. If an identical search is repeated within this window, the cached response is returned immediately with zero additional API calls. The cache holds a maximum of 100 entries; expired entries are purged automatically. A separate channel-info cache (maximum 200 entries, same 30-minute TTL) stores channel metadata fetched via channels.list.

Quota Optimisation

Technique	Detail	Quota Saving
In-Memory Cache (30 min TTL)	Identical searches served from cache — no API call made.	Saves 100+ units per repeated query
Batch API Requests	Up to 50 video IDs or channel IDs per request to videos.list / channels.list	1 unit for 50 items vs. 50 units individually
On-Demand Only	No background polling, auto-refresh, or prefetching.	Zero idle quota consumption
API Key Rotation	Multiple API keys supported; automatic rotation on 403/quota-exceeded responses.	Prevents service interruption at quota limit

Compliance Confirmation

- **Read-Only Access:** All API calls use GET requests only. No YouTube resources are created, updated, or deleted.
- **Official API Only:** All data is retrieved exclusively via the YouTube Data API v3 REST endpoints. No scraping or unofficial interfaces.
- **No Unauthorised Redistribution:** API data is displayed only within the Verdent application interface and is never sold or redistributed.
- **Attribution:** All video and channel data is attributed to YouTube/Google in the application UI footer.
- **Quota Compliance:** Daily quota is monitored via /api/quota. The app warns users when usage exceeds 80% and respects the 10,000-unit daily limit.
- **No Personal Data:** No OAuth authentication is performed. No user viewing history, subscriptions, or personally identifiable information is accessed.
- **Data Retention:** API responses are cached in-memory for a maximum of 30 minutes and are never written to persistent storage.
- **YouTube Attribution:** Every video card provides a direct link to the original video on YouTube.com.

This report was generated programmatically using the Verdent compliance reporting system. Screenshots were captured from the live application using Playwright (headless Chromium, viewport 1280×800). For questions, contact developer@verdent.app.